Lecture 5: Hypothesis testing

Economics 527: Econometric Methods

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Hypotheses and errors

Estimator and parameter set

- Normally distributed estimator with known variance ω^2:

$$\underset{1\times 1}{\hat{\theta}} \sim N(\underset{1\times 1}{\theta}, \underset{1\times 1}{\omega^2}), \qquad \frac{\hat{\theta}-\theta}{\omega} = Z \sim N(0,1)$$

- Regression example, conditional on X:

$$\theta = \beta_1,\ \hat{\theta} = \hat{\beta}_1,\ \omega^2 = \sigma^2/(X_1^\top M_2 X_1).$$

- Parameter set, for example $\Theta = \mathbb{R}$:

$$\theta \in \Theta$$

- Two disjoint parts:

$$\Theta = \Theta_0 \cup \Theta_1$$
$$\Theta_0 \cap \Theta_1 = \emptyset$$

- Null and alternative hypotheses:

$$H_0 : \theta \in \Theta_0 \qquad \text{vs} \qquad H_1 : \theta \in \Theta_1$$

Examples of Θ_0 and Θ_1

- Point null, $\Theta = \mathbb{R}$:

$$\Theta_0 = \{0\}$$
$$\Theta_1 = \mathbb{R} \setminus \{0\}$$
$$\iff \quad H_0 : \theta = 0 \quad \text{vs} \quad H_1 : \theta \neq 0$$

- More generally, for a **known** θ_0:

$$\Theta_0 = \{\theta_0\}$$
$$\Theta_1 = \mathbb{R} \setminus \{\theta_0\}$$
$$\iff \quad H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \theta \neq \theta_0$$

- One-sided null:

$$\Theta_0 = [\,\theta_0,\ +\infty)$$
$$\Theta_1 = (-\infty,\ \theta_0)$$
$$\iff \quad H_0 : \theta \geq \theta_0 \quad \text{vs} \quad H_1 : \theta < \theta_0$$

- **Definition.** Simple hypothesis: a single-point set; otherwise composite:

$$\Theta_0 = \{\theta_0\} : \text{ simple}, \qquad \Theta_0 = [\,0,\ +\infty) : \text{ composite}$$

Test statistic and decision rule

- Test statistic T: a function of the data, random, with range S:

$$\begin{aligned} &T \in S \\ &S = S_A \cup S_R, \qquad S_A \cap S_R = \emptyset \end{aligned}$$

- S_A: acceptance region; S_R: rejection region.
- Decision rule:

$$\begin{aligned} T \in S_A \quad &\Longrightarrow \quad \text{decide in favor of } \theta \in \Theta_0 \qquad (\text{fail to reject } H_0) \\ T \in S_R \quad &\Longrightarrow \quad \text{decide in favor of } \theta \in \Theta_1 \qquad (\text{reject } H_0) \end{aligned}$$

Type I and Type II errors

Truth against decision:

Truth	Decision H_0	Decision H_1
H_0	correct	Type I error
H_1	Type II error	correct

- Probabilities of the errors:

$$\begin{aligned} \mathrm{P}(\text{Type I error}) &= \mathrm{P}(T \in S_R \mid \theta \in \Theta_0) \\ \mathrm{P}(\text{Type II error}) &= \mathrm{P}(T \notin S_R \mid \theta \in \Theta_1) \\ &= \mathrm{P}(T \in S_A \mid \theta \in \Theta_1) \end{aligned}$$

- To make P(Type I error) smaller:

$$\begin{aligned} & S_R \downarrow \\ \Longrightarrow \quad & S_A \uparrow \\ \Longrightarrow \quad & \mathrm{P}(\text{Type II error}) \uparrow \end{aligned}$$

- We can't make both error probabilities small at the same time.

Valid test, size, significance level

- **Definition.** Valid test, with α chosen in advance:

$$\begin{aligned} \mathrm{P}(\text{Type I error}) &= \mathrm{P}(\text{falsely deciding in favour of } H_1 \mid H_0 \text{ true}) \\ &\leq \alpha \end{aligned}$$

- Composite Θ_0, supremum of the Type I error probabilities:

$$\sup_{\theta \in \Theta_0} \mathrm{P}(T \in S_R \mid \theta) \leq \alpha$$

- Size α test:

$$\sup_{\theta \in \Theta_0} \mathrm{P}(T \in S_R \mid \theta) = \alpha$$

- Significance level α: small, for example 0.01, 0.05, 0.1.

Two-sided test

Point null against a two-sided alternative

- Hypotheses, for example $\theta_0 = 0$:

$$\underbrace{H_0 : \theta = \theta_0}_{\text{simple}} \qquad \text{vs} \qquad \underbrace{H_1 : \theta \neq \theta_0}_{\text{composite}}$$

- Three objects:

$$\begin{aligned} \theta &: \text{ true value, unknown, fixed} \\ \hat{\theta} &: \text{ estimator, data, random} \\ \theta_0 &: \text{ chosen by you, known, fixed} \end{aligned}$$

- Regression example, $\theta = \beta_1$ and $\theta_0 = 0$:

$$H_0 : \beta_1 = 0 \qquad \text{vs} \qquad H_1 : \beta_1 \neq 0$$

Test statistic $T(\theta_0)$

- Distance between the data and H_0:

$$T(\theta_0) = \left| \frac{\hat{\theta} - \theta_0}{\omega} \right|$$

- Reject H_0 when:

$$\begin{aligned} & T(\theta_0) \text{ is large} \\ \iff \quad & T(\theta_0) > c \end{aligned}$$

- Regions, $c > 0$:

$$\begin{aligned} S_R &= (c,\ +\infty) \\ S_A &= [\,0,\ c\,] \\ S &= S_A \cup S_R = [\,0,\ +\infty) \end{aligned}$$

- Choose c for size control:

$$\text{P(Type I error)} \leq \alpha$$

Choice of c

- Size α: set the Type I error probability at $\theta = \theta_0$ to α:

$$\begin{aligned}
\alpha &= \mathrm{P}(\text{Type I error}) \\
&= \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ true}) \\
&= \mathrm{P}(T(\theta_0) > c \mid \theta = \theta_0) \\
&= \mathrm{P}\Big(\Big|\frac{\hat\theta - \theta_0}{\omega}\Big| > c \;\Big|\; \theta = \theta_0\Big) \\
&\overset{\theta=\theta_0}{=} \mathrm{P}\Big(\Big| \underbrace{\frac{\hat\theta - \theta}{\omega}}_{=Z\sim N(0,1)} \Big| > c\Big) \\
&= \mathrm{P}(|Z| > c) \\
&= \mathrm{P}(\underbrace{\{Z > c\}}_{A} \text{ or } \underbrace{\{-Z > c\}}_{B}) \\
&= \mathrm{P}(Z > c) + \mathrm{P}(Z < -c) \\
&= 2\,\mathrm{P}(Z > c) \\
\implies \quad \mathrm{P}(Z > c) &= \frac{\alpha}{2} \\
\implies \quad \mathrm{P}(Z \le c) &= 1 - \frac{\alpha}{2} \\
\implies \quad c &= z_{1-\alpha/2} \qquad (\mathrm{P}(Z \le z_\tau) = \tau)
\end{aligned}$$

Size α two-sided test

- Size α test of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$:

$$\boxed{\text{Reject } H_0 \;\text{ if }\; \left|\frac{\hat\theta - \theta_0}{\omega}\right| > z_{1-\alpha/2}}$$

- For $\alpha = 0.05$:

$$\begin{aligned}
& z_{0.975} \approx 1.96 \approx 2 \\
\implies \quad & \text{reject if } \Big|\frac{\hat\theta - \theta_0}{\omega}\Big| \gtrsim 2 \\
\iff \quad & \text{reject if } |\hat\theta - \theta_0| \gtrsim 2 \underbrace{\omega}_{\text{std. dev.}}
\end{aligned}$$

- Compare with: Reject if $|\hat\theta - \theta_0| > 5\,\omega$:

$$\mathrm{P}(Z > 5) \approx 3 \cdot 10^{-7}, \qquad \alpha = \mathrm{P}(|Z| > 5) \approx 6 \cdot 10^{-7}$$

Significance at level α

- **Definition.** Significance at level α, by a size α test:

$$\begin{aligned}
\Big|\frac{\hat\theta}{\omega}\Big| > z_{1-\alpha/2} \quad &\implies \quad H_0 : \theta = 0 \;\text{ rejected in favour of }\; H_1 : \theta \neq 0 \\
&\implies \quad \hat\theta \;\text{ significant at level } \alpha
\end{aligned}$$

Test and confidence interval

- Solve for θ_0:

$$\begin{aligned}
H_0 \text{ rejected} &\iff \left|\frac{\hat{\theta}-\theta_0}{\omega}\right| > z_{1-\alpha/2} \\
H_0 \text{ not rejected} &\iff \left|\frac{\hat{\theta}-\theta_0}{\omega}\right| \le z_{1-\alpha/2} \\
&\iff -z_{1-\alpha/2} \le \frac{\hat{\theta}-\theta_0}{\omega} \le z_{1-\alpha/2} \\
&\iff -z_{1-\alpha/2}\,\omega \le \hat{\theta}-\theta_0 \le z_{1-\alpha/2}\,\omega \\
&\iff z_{1-\alpha/2}\,\omega \ge \theta_0-\hat{\theta} \ge -z_{1-\alpha/2}\,\omega \\
&\iff \hat{\theta}+z_{1-\alpha/2}\,\omega \ge \theta_0 \ge \hat{\theta}-z_{1-\alpha/2}\,\omega \\
&\iff \hat{\theta}-z_{1-\alpha/2}\,\omega \le \theta_0 \le \hat{\theta}+z_{1-\alpha/2}\,\omega \\
&\iff \theta_0 \in CI_{1-\alpha} = [\hat{\theta}-z_{1-\alpha/2}\,\omega,\ \hat{\theta}+z_{1-\alpha/2}\,\omega]
\end{aligned}$$

- Equivalent statement of the test:

$$\boxed{\text{Reject } H_0 : \theta = \theta_0 \ \text{ if } \ \theta_0 \notin CI_{1-\alpha}}$$

$CI_{1-\alpha}$ as a set of null values

- All θ_0 the test does not reject:

$$\begin{aligned}
CI_{1-\alpha} &= \{\theta_0 : T(\theta_0) \le z_{1-\alpha/2}\} \\
&= \left\{\theta_0 : \left|\frac{\hat{\theta}-\theta_0}{\omega}\right| \le z_{1-\alpha/2}\right\} \\
&= \{\theta_0 : \text{a size } \alpha \text{ test does not reject } H_0 : \theta = \theta_0\}
\end{aligned}$$

- Suppose in your data:

$$CI_{0.95} = [\,0.04,\ 0.06\,]$$

Power

Noise and signal

- Type II error:

$$\text{P(Type II error)} = \text{P(fail to reject } H_0 \mid \theta \ne \theta_0)$$

- Split $(\hat{\theta}-\theta_0)/\omega$, under the true θ:

$$\begin{aligned}
\underbrace{\frac{\hat{\theta}-\theta_0}{\omega}}_{\text{data}-H_0} &= \underbrace{\frac{\hat{\theta}-\theta}{\omega}}_{=Z\sim N(0,1),\ \text{noise}} + \underbrace{\frac{\theta-\theta_0}{\omega}}_{\text{truth}-H_0,\ \text{signal}} \\
&= Z + \frac{\theta-\theta_0}{\omega} \\
&\sim N\Big(\frac{\theta-\theta_0}{\omega},\ 1\Big)
\end{aligned}$$

Power function $\pi(\theta)$

- Probability of rejecting H_0, as a function of the true θ:

$$\begin{aligned}\pi(\theta) &= \mathrm{P}(\text{reject } H_0 \mid \theta) \\ &= \mathrm{P}\Big(\Big|\frac{\hat{\theta}-\theta_0}{\omega}\Big| > z_{1-\alpha/2} \;\Big|\; \theta\Big) \\ &= \mathrm{P}\Big(\Big|\frac{\hat{\theta}-\theta}{\omega} + \frac{\theta-\theta_0}{\omega}\Big| > z_{1-\alpha/2} \;\Big|\; \theta\Big) \\ &= \mathrm{P}\Big(\Big|\underset{N(0,1)}{Z} + \frac{\theta-\theta_0}{\omega}\Big| > z_{1-\alpha/2}\Big)\end{aligned}$$

- Known: θ_0, ω, $z_{1-\alpha/2}$, the distribution of Z.
- Unknown: $\theta \in \mathbb{R}$.

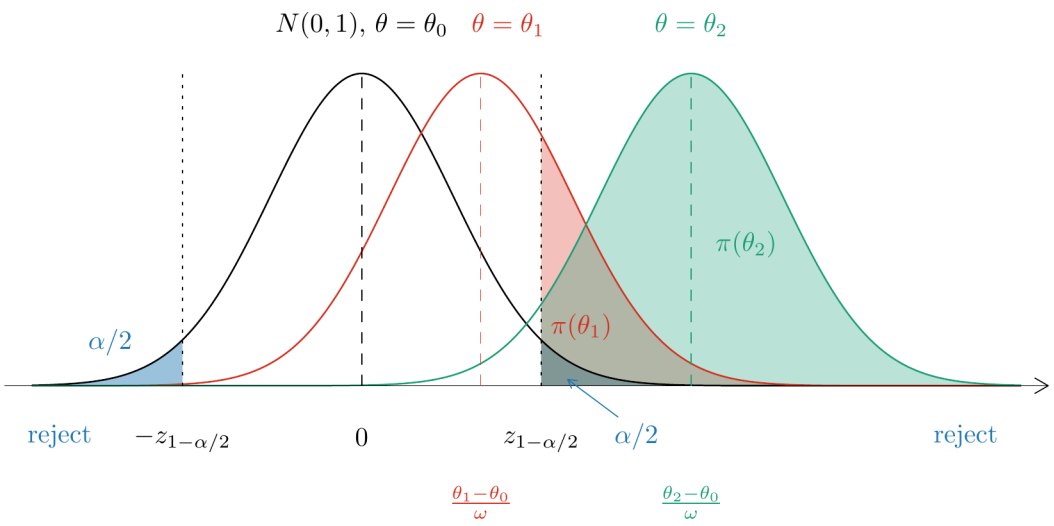
Type II error probability

- Type II error probability depends on θ: For $\theta \neq \theta_0$,

$$\begin{aligned}\mathrm{P}(\text{Type II error} \mid \theta) &= \mathrm{P}(\text{fail to reject } H_0 \mid \theta) \\ &= 1 - \mathrm{P}(\text{reject } H_0 \mid \theta) \\ &= 1 - \pi(\theta).\end{aligned}$$

Distribution of $(\hat{\theta}-\theta_0)/\omega$ under H_0 and H_1

- Density of $(\hat{\theta}-\theta_0)/\omega$, with $\theta_0 < \theta_1 < \theta_2$:

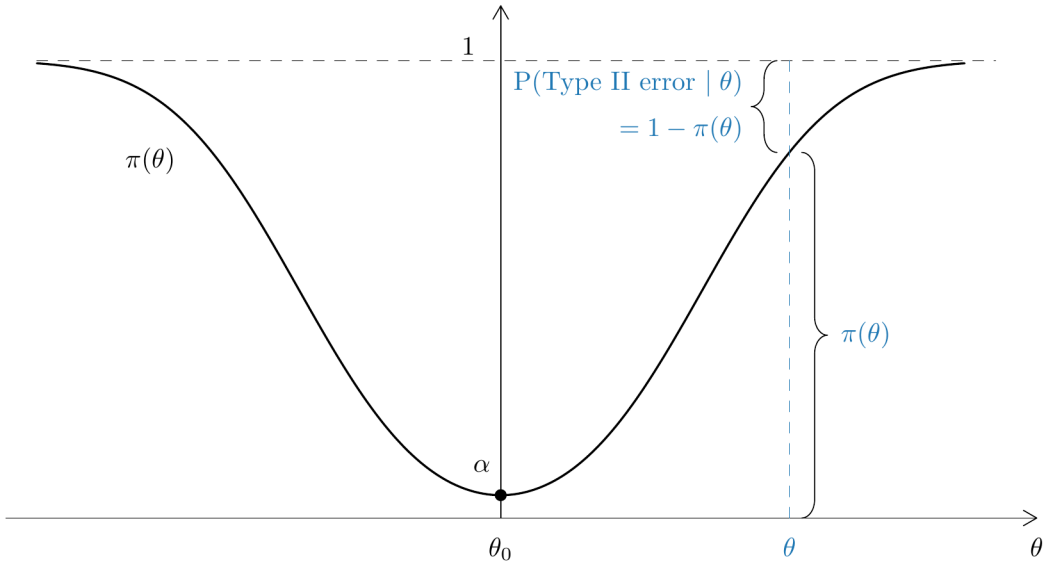


- Under $\theta \neq \theta_0$, more mass in the rejection region:

$$\mathrm{P}\Big(\Big|\frac{\hat{\theta}-\theta_0}{\omega}\Big| > z_{1-\alpha/2} \;\Big|\; \theta\Big) > \alpha$$

Power curve

- $\pi(\theta)$ against θ: minimum α at θ_0, approaches 1 far from θ_0:



- Largest Type II error probability, as $\theta \to \theta_0$:

$$\begin{aligned}\sup_{\theta \neq \theta_0} \mathrm{P}(\text{Type II error} \mid \theta) &= \sup_{\theta \neq \theta_0} \mathrm{P}\Big(\Big|\frac{\hat{\theta} - \theta_0}{\omega}\Big| \leq z_{1-\alpha/2} \;\Big|\; \theta\Big) \\ &= \sup_{\theta \neq \theta_0} \mathrm{P}\Big(\Big|Z + \frac{\theta - \theta_0}{\omega}\Big| \leq z_{1-\alpha/2}\Big) \\ &= 1 - \alpha\end{aligned}$$

Maximum against supremum

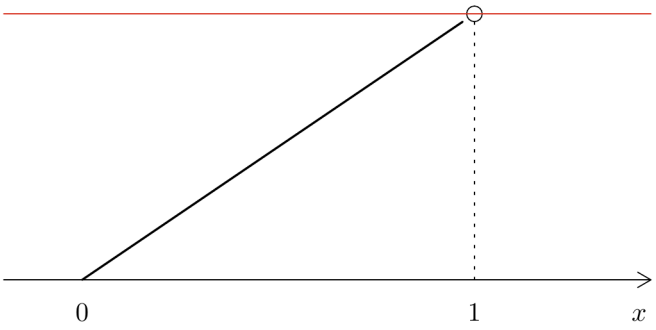
- Closed interval:

$$\max_{x \in [0,1]} x = 1$$

- Half-open interval:

$$\max_{x \in [0,1)} x \text{ does not exist}$$

$$\sup_{x \in [0,1)} x = 1$$



- **Definition.** Supremum:

$$\sup_{x \in \mathcal{X}} f(x) = \text{smallest upper bound of } \{f(x) : x \in \mathcal{X}\}$$

- P(Type II error | θ) over $\theta \neq \theta_0$: no maximum.

Example: $\hat{\theta} \sim N(\theta,\ 9)$

- $H_0 : \theta = 0$ vs $H_1 : \theta \neq 0$, $\alpha = 0.05$, $\omega = 3$:

$$\text{Reject } H_0 \ \text{ if } \ \Big|\frac{\hat{\theta}}{3}\Big| > 1.96$$

- Power function:

$$\begin{aligned}
\pi(\theta) &= \mathrm{P}\Big(\Big|\frac{\hat{\theta}}{3}\Big| > 1.96 \ \Big|\ \theta\Big) \\
&= \mathrm{P}\Big(\Big|\frac{\hat{\theta}-\theta}{3} + \frac{\theta}{3}\Big| > 1.96\Big) \\
&= \mathrm{P}\Big(\Big|Z + \frac{\theta}{3}\Big| > 1.96\Big) \\
&= \mathrm{P}\Big(Z + \frac{\theta}{3} > 1.96 \ \text{ or } \ Z + \frac{\theta}{3} < -1.96\Big) \\
&= \mathrm{P}\Big(Z > 1.96 - \frac{\theta}{3}\Big) + \mathrm{P}\Big(Z < -1.96 - \frac{\theta}{3}\Big)
\end{aligned}$$

Example, power at $\theta = 3$

- Substitute $\theta = 3$:

$$\begin{aligned}
\pi(3) &= \mathrm{P}\Big(Z > 1.96 - \frac{3}{3}\Big) + \mathrm{P}\Big(Z < -1.96 - \frac{3}{3}\Big) \\
&= \mathrm{P}(Z > 0.96) + \mathrm{P}(Z < -2.96) \\
&\approx \underbrace{\mathrm{P}(Z \leq -0.96)}_{\approx 0.16853} + 0.00154 \\
&\approx 0.17
\end{aligned}$$

- $\pi(\theta)$ at four values of θ:

$$\pi(\theta) \approx \begin{cases} 0.063 & \theta = -1 \\ 0.05 & \theta = 0 \\ 0.063 & \theta = 1 \\ 0.17 & \theta = 3 \end{cases}$$

Failing to reject $\beta_1 = 0$

- Regression, $\theta = \beta_1$, $\omega^2 = \sigma^2/(X_1^\top M_2 X_1)$:

$$Y = \beta_1 X_1 + X_2\beta_2 + U, \qquad H_0 : \beta_1 = 0 \quad \text{vs} \quad H_1 : \beta_1 \neq 0$$

- Suppose H_0 is not rejected:

$$\Big|\frac{\hat{\beta}_1}{\omega}\Big| \leq z_{1-\alpha/2} \qquad \overset{?}{\Longrightarrow} \qquad \beta_1 = 0$$

- Type II error probability near $1 - \alpha$ for θ near θ_0.
- $T(\theta_0) > z_{1-\alpha/2}$: strong evidence that $\theta \neq \theta_0$.
- **Claim.** $T(\theta_0) \leq z_{1-\alpha/2}$: not strong evidence in favour of θ_0.

Minimum detectable deviation

- Detectable θ: $\pi(\theta) \geq \beta$, for example $\beta = 0.85$ or 0.5.
- Smallest detectable $\bar{\theta} > \theta_0$ solves:

$$\mathrm{P}\Big(\Big|Z + \frac{\bar{\theta} - \theta_0}{\omega}\Big| > z_{1-\alpha/2}\Big) = \beta, \qquad \theta_0,\ \omega,\ z_{1-\alpha/2},\ \beta \ \text{ known}$$

- $|\bar{\theta} - \theta_0|$: the minimum detectable deviation.

More precise estimator, more power

- Smaller variance:

$$\begin{aligned} & \hat{\theta}^* \sim N(\theta,\ \omega_*^2) \\ & \omega_*^2 < \omega^2 \\ \Longrightarrow \quad & \omega_* < \omega \end{aligned}$$

- Both tests valid:

$$\underbrace{\Big|\frac{\hat{\theta} - \theta_0}{\omega}\Big| > z_{1-\alpha/2}}_{\text{test with } \hat{\theta}} \qquad \text{or} \qquad \underbrace{\Big|\frac{\hat{\theta}^* - \theta_0}{\omega_*}\Big| > z_{1-\alpha/2}}_{\text{test with } \hat{\theta}^*}$$

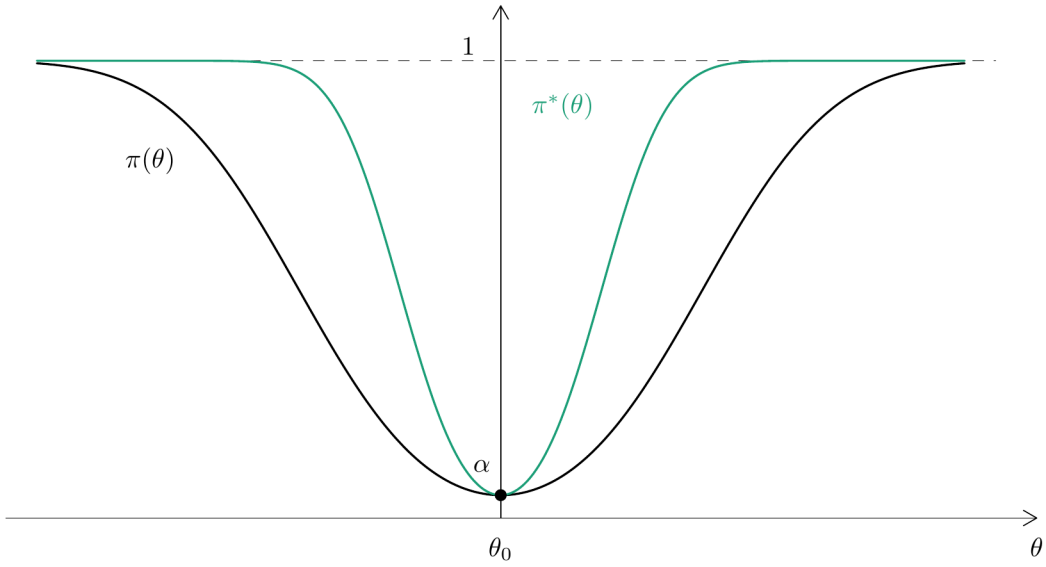
- Suppose $\theta > \theta_0$, larger signal with $\hat{\theta}^*$:

$$\begin{aligned} & \frac{\hat{\theta} - \theta_0}{\omega} = Z + \frac{\theta - \theta_0}{\omega} \sim N\Big(\frac{\theta - \theta_0}{\omega},\ 1\Big) \\ & \frac{\hat{\theta}^* - \theta_0}{\omega_*} \sim N\Big(\frac{\theta - \theta_0}{\omega_*},\ 1\Big) \\ \omega_* < \omega \quad \Longrightarrow \quad & \frac{\theta - \theta_0}{\omega} < \frac{\theta - \theta_0}{\omega_*} \end{aligned}$$

- Power functions:

$$\begin{aligned} \pi(\theta) &= \mathrm{P}\Big(\Big|\frac{\hat{\theta} - \theta_0}{\omega}\Big| > z_{1-\alpha/2} \,\Big|\, \theta\Big) \\ \pi^*(\theta) &= \mathrm{P}\Big(\Big|\frac{\hat{\theta}^* - \theta_0}{\omega_*}\Big| > z_{1-\alpha/2} \,\Big|\, \theta\Big) \end{aligned}$$

- Test with $\hat{\theta}^*$: more powerful, $\pi^*(\theta) \geq \pi(\theta)$.



Test that ignores the data

- $V \sim \text{Uniform}(0, 1)$, density and CDF:

$$\begin{aligned} f(v) &= \mathbf{1}\{0 < v < 1\} \\ &= \begin{cases} 1 & 0 < v < 1 \\ 0 & \text{otherwise} \end{cases} \\ F(v) &= \begin{cases} 0 & v \le 0 \\ v & 0 < v < 1 \\ 1 & v \ge 1 \end{cases} \end{aligned}$$

- For $0 < x < 1$:

$$\begin{aligned} \mathrm{P}(V < x) &= \mathrm{P}(V \le x) \qquad \text{(continuous)} \\ &= F(x) \\ &= \int_0^x f(v)\, dv \\ &= \int_0^x 1\, dv \\ &= x \end{aligned}$$

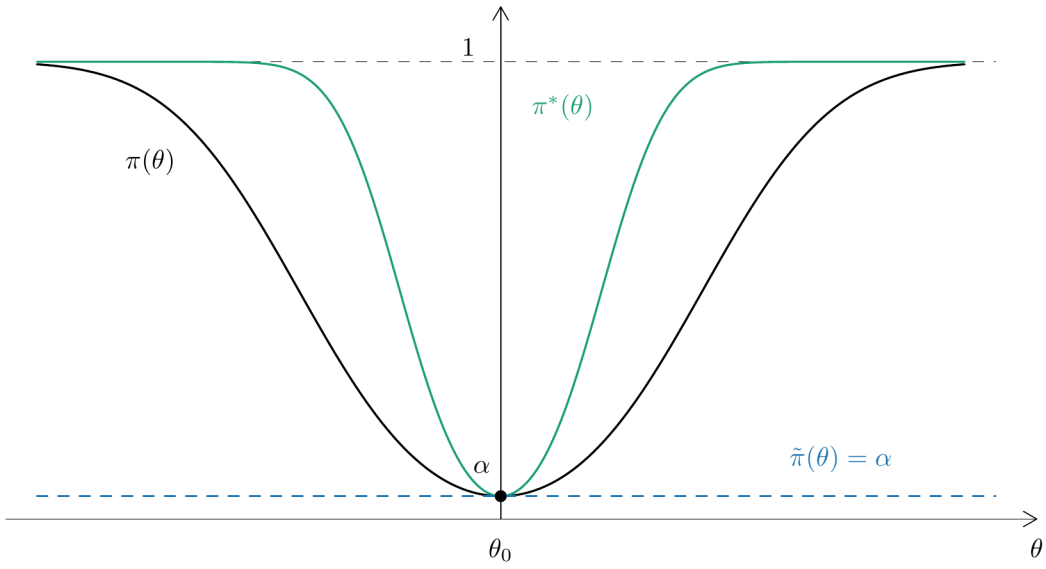
- Test:

1. Throw $\hat{\theta}$ away.
2. Draw $V \sim \text{Uniform}(0, 1)$.
3. Reject $H_0 : \theta = \theta_0$ if $V < \alpha$.

- V does not depend on θ:

$$\begin{aligned} \mathrm{P}(\text{Type I error} \mid \theta = \theta_0) &= \mathrm{P}(V < \alpha) \\ &= \alpha \\ \tilde{\pi}(\theta) &= \mathrm{P}(\text{reject } H_0 \mid \theta) \\ &= \mathrm{P}(V < \alpha) \\ &= \alpha \qquad \text{for every } \theta \end{aligned}$$

- Power α: trivial.



One-sided tests

One-sided hypotheses

- One-sided null and alternative, for example $\theta_0 = 0$:

$$\underbrace{H_0 : \theta \leq \theta_0}_{\text{composite}} \qquad \text{vs} \qquad \underbrace{H_1 : \theta > \theta_0}_{\text{composite}}$$

- Test: reject H_0 in favour of H_1 if:

$$\frac{\hat{\theta} - \theta_0}{\omega} > c$$

- Type I error probability for each $\theta \in \Theta_0$, and its supremum:

$$\mathrm{P}(\text{Type I error} \mid \theta) = \mathrm{P}\Big(\frac{\hat{\theta} - \theta_0}{\omega} > c \,\Big|\, \theta\Big), \qquad \theta \leq \theta_0$$

$$\sup_{\theta \leq \theta_0} \mathrm{P}(\text{Type I error} \mid \theta) \leq \alpha$$

- Split $(\hat{\theta} - \theta_0)/\omega$:

$$\frac{\hat{\theta} - \theta_0}{\omega} = \underbrace{\frac{\hat{\theta} - \theta}{\omega}}_{= Z} + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\leq 0 \text{ for } \theta \leq \theta_0}$$

Choice of c, one-sided

- Supremum over Θ_0, attained at $\theta = \theta_0$:

$$\begin{aligned}
\alpha &= \sup_{\theta \le \theta_0} \mathrm{P}(\text{Type I error} \mid \theta) \\
&= \sup_{\theta \le \theta_0} \mathrm{P}(\text{reject } H_0 \mid \theta) \\
&= \sup_{\theta \le \theta_0} \mathrm{P}\Big(\frac{\hat\theta - \theta_0}{\omega} > c \,\Big|\, \theta \Big) \\
&= \sup_{\theta \le \theta_0} \mathrm{P}\Big(\underbrace{\frac{\hat\theta - \theta}{\omega}}_{= Z \sim N(0,1)} + \frac{\theta - \theta_0}{\omega} > c \,\Big|\, \theta \Big) \\
&= \sup_{\theta \le \theta_0} \mathrm{P}\Big(Z + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\le 0,\ \text{largest at } \theta = \theta_0} > c \Big) \\
&= \mathrm{P}(Z > c) \\
\implies \quad \mathrm{P}(Z \le c) &= 1 - \alpha \\
\implies \quad c &= z_{1-\alpha}
\end{aligned}$$

One-sided test and interval

- Size α test of $H_0 : \theta \le \theta_0$ against $H_1 : \theta > \theta_0$:

$$\boxed{\text{Reject } H_0 \ \text{ if } \ \frac{\hat\theta - \theta_0}{\omega} > z_{1-\alpha}}$$

- One-sided interval, all θ_0 the test does not reject:

$$\begin{aligned}
CI^1_{1-\alpha} &= \Big\{ \theta_0 : \frac{\hat\theta - \theta_0}{\omega} \le z_{1-\alpha} \Big\} \\
&= \{ \theta_0 : \theta_0 \ge \hat\theta - z_{1-\alpha}\, \omega \} \\
&= [\hat\theta - z_{1-\alpha}\, \omega,\ +\infty)
\end{aligned}$$

Mirror case: $H_0 : \theta \ge \theta_0$

- Reject $H_0 : \theta \ge \theta_0$ in favour of $H_1 : \theta < \theta_0$ if, for some $c > 0$:

$$\frac{\hat\theta - \theta_0}{\omega} < -c$$

- Choose c from the supremum over $\theta \ge \theta_0$:

$$\begin{aligned}
\alpha &\ge \mathrm{P}(\text{Type I error} \mid \theta) \qquad (\theta \ge \theta_0) \\
&= \mathrm{P}\Big(\frac{\hat\theta - \theta_0}{\omega} < -c \,\Big|\, \theta \Big) \\
&= \mathrm{P}\Big(Z + \frac{\theta - \theta_0}{\omega} < -c \Big) \\
\alpha &= \sup_{\theta \ge \theta_0} \mathrm{P}\Big(Z + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\ge 0} < -c \Big) \qquad (\text{size } \alpha) \\
&= \mathrm{P}(Z < -c) \qquad (\theta = \theta_0) \\
\implies \quad -c &= z_\alpha \\
\implies \quad c &= z_{1-\alpha}
\end{aligned}$$

- Test:

$$\boxed{\text{Reject } H_0 : \theta \geq \theta_0 \text{ if } \frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-\alpha}}$$

Power through Φ

- $\Phi(x) = \mathrm{P}(Z \leq x)$, the CDF of $N(0,1)$.
- Left-tailed test of $H_0 : \theta \geq \theta_0$:

$$\begin{aligned}
\pi(\theta) &= \mathrm{P}\Big(\frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-\alpha} \,\Big|\, \theta\Big) \\
&= \mathrm{P}\Big(Z + \frac{\theta - \theta_0}{\omega} < -z_{1-\alpha}\Big) \\
&= \mathrm{P}\Big(Z < -z_{1-\alpha} - \frac{\theta - \theta_0}{\omega}\Big) \\
&= \Phi\Big(\frac{\theta_0 - \theta}{\omega} - z_{1-\alpha}\Big)
\end{aligned}$$

Two tests for $H_0 : \theta = \theta_0$

- Against $H_1 : \theta \neq \theta_0$, both of size α:

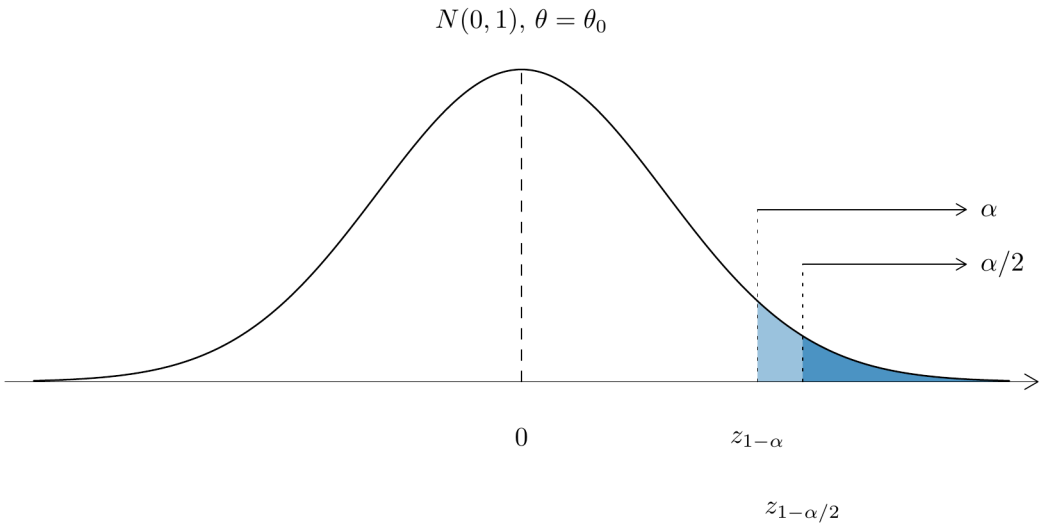
$$\begin{aligned}
&\text{Test 1 (one-sided)}: \quad \text{reject } H_0 \text{ if } \frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha} \\
&\text{Test 2 (two-sided)}: \quad \text{reject } H_0 \text{ if } \Big|\frac{\hat{\theta} - \theta_0}{\omega}\Big| > z_{1-\alpha/2}
\end{aligned}$$

- Type I error of Test 1:

$$\begin{aligned}
\mathrm{P}(\text{Type I error of Test 1}) &= \mathrm{P}\Big(\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha} \,\Big|\, \theta = \theta_0\Big) \\
&\overset{\theta=\theta_0}{=} \mathrm{P}\Big(\frac{\hat{\theta} - \theta}{\omega} > z_{1-\alpha}\Big) \\
&= \mathrm{P}(Z > z_{1-\alpha}) \\
&= \alpha
\end{aligned}$$

- Critical values:

$$\begin{aligned}
& z_{1-\alpha} < z_{1-\alpha/2} \\
\implies \quad & -z_{1-\alpha/2} < -z_{1-\alpha}
\end{aligned}$$



Power of Test 1 and Test 2

- Power of Test 1:

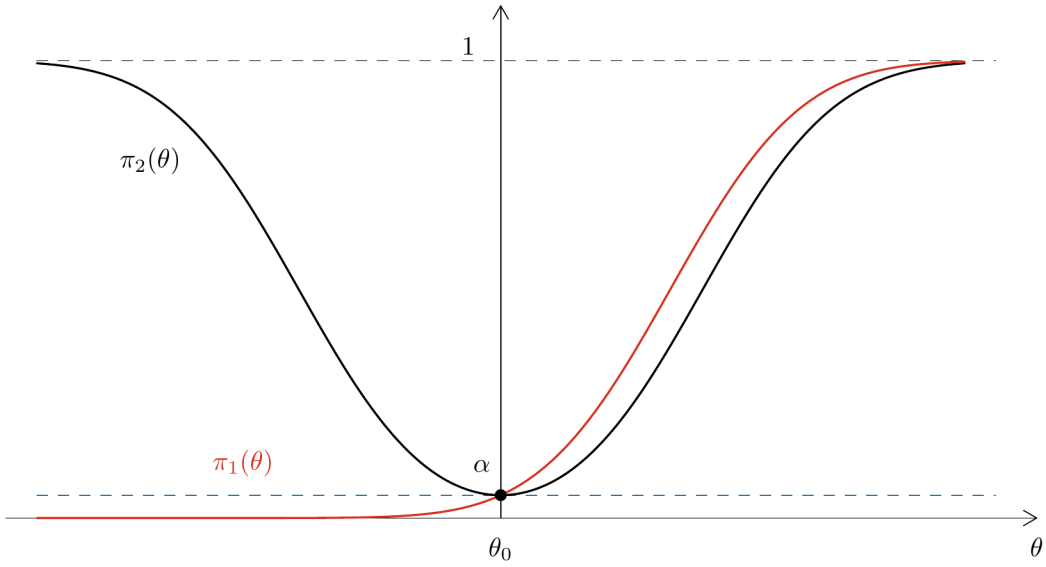
$$\begin{aligned}\pi_1(\theta) &= \mathrm{P}\Big(\frac{\hat{\theta}-\theta_0}{\omega} > z_{1-\alpha} \,\Big|\, \theta\Big) \\ &= \mathrm{P}\Big(Z + \frac{\theta-\theta_0}{\omega} > z_{1-\alpha}\Big)\end{aligned}$$

- Power of Test 2:

$$\begin{aligned}\pi_2(\theta) &= \mathrm{P}\Big(\Big|\frac{\hat{\theta}-\theta_0}{\omega}\Big| > z_{1-\alpha/2} \,\Big|\, \theta\Big) \\ &= \mathrm{P}\Big(\Big|Z + \frac{\theta-\theta_0}{\omega}\Big| > z_{1-\alpha/2}\Big) \\ &= \mathrm{P}\Big(Z + \frac{\theta-\theta_0}{\omega} < -z_{1-\alpha/2}\Big) + \mathrm{P}\Big(Z + \frac{\theta-\theta_0}{\omega} > z_{1-\alpha/2}\Big)\end{aligned}$$

One-sided against two-sided power

- $\pi_1(\theta)$ increasing, $\pi_2(\theta)$ U-shaped:



- For $\theta > \theta_0$, Test 1 is more powerful; compare the right tails:

$$\left|\frac{\hat{\theta}-\theta_0}{\omega}\right| = \frac{\hat{\theta}-\theta_0}{\omega} > z_{1-\alpha/2} \qquad \left(\frac{\hat{\theta}-\theta_0}{\omega} > 0\right)$$
$$\implies \quad \frac{\hat{\theta}-\theta_0}{\omega} > z_{1-\alpha}$$

- For $\theta < \theta_0$, Test 2 is more powerful:

$$\frac{\hat{\theta}-\theta_0}{\omega} = Z + \underbrace{\frac{\theta-\theta_0}{\omega}}_{<0} > z_{1-\alpha} \qquad \text{(Test 1 rejects)}$$
$$\begin{aligned}\pi_1(\theta) &= \mathrm{P}\left(Z + \frac{\theta-\theta_0}{\omega} > z_{1-\alpha}\right) \\ &= \mathrm{P}\left(Z > z_{1-\alpha} - \frac{\theta-\theta_0}{\omega}\right) \\ &< \mathrm{P}(Z > z_{1-\alpha}) \\ &= \alpha\end{aligned}$$

- No uniformly most powerful test against $H_1 : \theta \neq \theta_0$.
- Test 1 biased against $H_1 : \theta \neq \theta_0$.
- Largest Type II error probability of Test 1:

$$\text{against } H_1 : \theta > \theta_0 : \quad \sup_{\theta>\theta_0} (1 - \pi_1(\theta)) = 1 - \alpha$$
$$\text{against } H_1 : \theta \neq \theta_0 : \quad \sup_{\theta\neq\theta_0} (1 - \pi_1(\theta)) = 1$$

Two-sided test with unequal tails

- Test 3 of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$, reject H_0 when:

$$\frac{\hat{\theta}-\theta_0}{\omega} > z_{1-\alpha/3} \qquad \text{or} \qquad \frac{\hat{\theta}-\theta_0}{\omega} < -z_{1-2\alpha/3}$$

- Type I error probability, with $z_\tau = -z_{1-\tau}$:

$$\begin{aligned}
\mathrm{P}(\text{Type I error}) &= \mathrm{P}\Big(\frac{\hat\theta-\theta_0}{\omega} > z_{1-\alpha/3} \;\text{ or }\; \frac{\hat\theta-\theta_0}{\omega} < -z_{1-2\alpha/3} \;\Big|\; \theta=\theta_0\Big) \\
&= \mathrm{P}\Big(\frac{\hat\theta-\theta_0}{\omega} > z_{1-\alpha/3} \;\Big|\; \theta=\theta_0\Big) \\
&\qquad + \mathrm{P}\Big(\frac{\hat\theta-\theta_0}{\omega} < -z_{1-2\alpha/3} \;\Big|\; \theta=\theta_0\Big) \\
&\overset{\theta=\theta_0}{=} \mathrm{P}\Big(\frac{\hat\theta-\theta}{\omega} > z_{1-\alpha/3}\Big) + \mathrm{P}\Big(\frac{\hat\theta-\theta}{\omega} < -z_{1-2\alpha/3}\Big) \\
&= \underbrace{\mathrm{P}(Z > z_{1-\alpha/3})}_{=\alpha/3} + \underbrace{\mathrm{P}(Z < -z_{1-2\alpha/3})}_{=\mathrm{P}(Z<z_{2\alpha/3})=2\alpha/3} \\
&= \alpha
\end{aligned}$$

Power with unequal tails

- Power of Test 3:

$$\begin{aligned}
\pi_3(\theta) &= \mathrm{P}(\text{reject } H_0 \mid \theta) \\
&= \mathrm{P}\Big(\frac{\hat\theta-\theta_0}{\omega} > z_{1-\alpha/3} \;\Big|\; \theta\Big) + \mathrm{P}\Big(\frac{\hat\theta-\theta_0}{\omega} < -z_{1-2\alpha/3} \;\Big|\; \theta\Big) \\
&= \mathrm{P}\Big(Z + \frac{\theta-\theta_0}{\omega} > z_{1-\alpha/3}\Big) + \mathrm{P}\Big(Z + \frac{\theta-\theta_0}{\omega} < -z_{1-2\alpha/3}\Big) \\
&= \mathrm{P}\Big(Z > z_{1-\alpha/3} - \frac{\theta-\theta_0}{\omega}\Big) + \mathrm{P}\Big(Z < -z_{1-2\alpha/3} - \frac{\theta-\theta_0}{\omega}\Big)
\end{aligned}$$

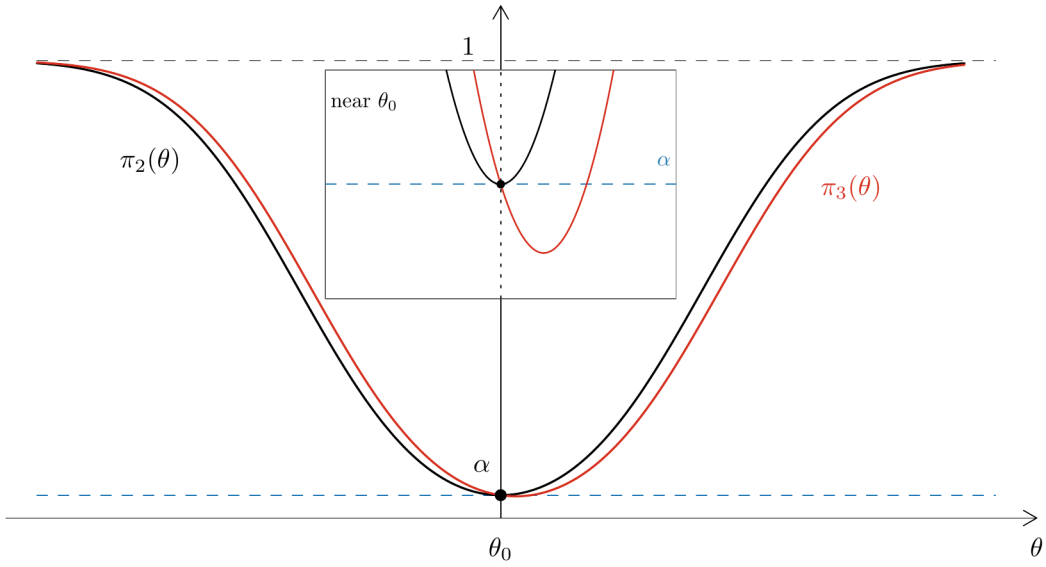
- At $\theta = \theta_0$:

$$\begin{aligned}
\pi_3(\theta_0) &= \mathrm{P}(Z > z_{1-\alpha/3}) + \mathrm{P}(Z < -z_{1-2\alpha/3}) \\
&= \frac{\alpha}{3} + \frac{2\alpha}{3} \\
&= \alpha
\end{aligned}$$

- For $\theta > \theta_0$, $(\theta-\theta_0)/\omega > 0$:

$$\pi_3(\theta) = \underbrace{\mathrm{P}\Big(Z > z_{1-\alpha/3} - \frac{\theta-\theta_0}{\omega}\Big)}_{>\alpha/3} + \underbrace{\mathrm{P}\Big(Z < -z_{1-2\alpha/3} - \frac{\theta-\theta_0}{\omega}\Big)}_{<2\alpha/3}$$

- $\pi_3(\theta)$ against $\pi_2(\theta)$, both α at θ_0:



- $\pi_3(\theta) < \alpha$ just right of θ_0: Test 3 biased against $H_1 : \theta \neq \theta_0$.

Strict inequality in the null

- $H_0 : \theta < \theta_0$ vs $H_1 : \theta \geq \theta_0$, same one-sided test:

$$\begin{aligned}\sup_{\theta<\theta_0} \mathrm{P}\Big(Z + \frac{\theta-\theta_0}{\omega} > z_{1-\alpha}\Big) &= \sup_{\theta\in(-\infty,\ \theta_0)} \mathrm{P}\Big(Z + \frac{\theta-\theta_0}{\omega} > z_{1-\alpha}\Big) \\ &= \mathrm{P}(Z > z_{1-\alpha}) \\ &= \alpha\end{aligned}$$

- $\theta_0 \in \Theta_1$: no power above α at θ_0.

Null $\theta \neq \theta_0$: not testable

- $H_0 : \theta \neq \theta_0$ vs $H_1 : \theta = \theta_0$; size control:

$$\sup_{\theta\neq\theta_0} \mathrm{P}(\text{reject } H_0 \mid \theta) \leq \alpha$$

- Reject if $|(\hat{\theta} - \theta_0)/\omega| < c$:

$$\begin{aligned}\alpha &\geq \sup_{\theta\neq\theta_0} \mathrm{P}\Big(\Big|\frac{\hat{\theta}-\theta_0}{\omega}\Big| < c \,\Big|\, \theta\Big) \\ &= \sup_{\theta\neq\theta_0} \mathrm{P}\Big(\Big|Z + \frac{\theta-\theta_0}{\omega}\Big| < c\Big) \\ &= \mathrm{P}(|Z| < c) \qquad (\theta \to \theta_0) \\ &= \mathrm{P}(\text{reject } H_0 \mid \theta = \theta_0)\end{aligned}$$

- Size α: no power above α at θ_0, not testable.

p-values

Two-sided p-value

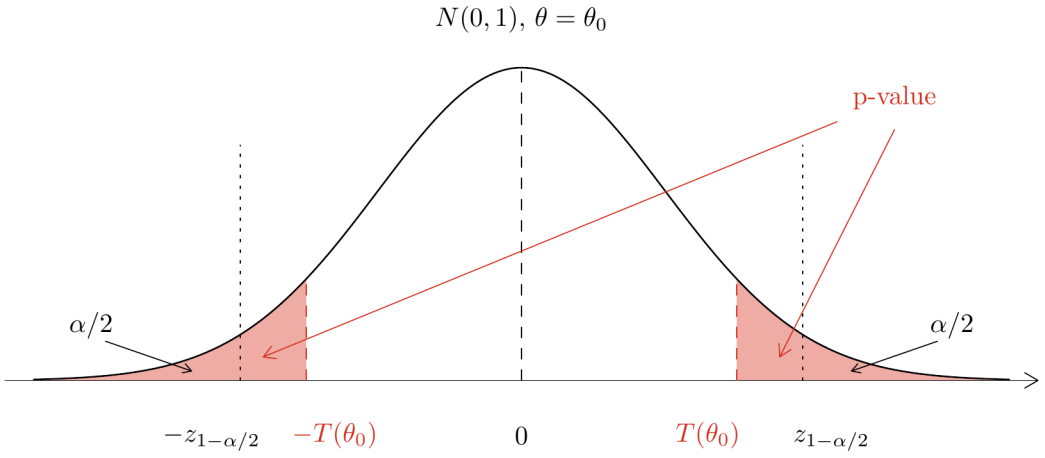
- Under $H_0 : \theta = \theta_0$:

$$\frac{\hat{\theta} - \theta_0}{\omega} \sim N(0,1)$$

- Test statistic:

$$T(\theta_0) = \left| \frac{\hat{\theta} - \theta_0}{\omega} \right|$$

- Two tails beyond $\pm T(\theta_0)$:



- Test:

Reject $H_0 : \theta = \theta_0$ if p-value $< \alpha$

- p-value:

$$\begin{aligned}
\text{p-value} &= \Phi(-T(\theta_0)) + 1 - \Phi(T(\theta_0)) \\
&= 2(1 - \Phi(T(\theta_0))) \qquad (\Phi(-x) = 1 - \Phi(x)) \\
&= \boxed{2\Big(1 - \Phi\Big(\Big|\frac{\hat{\theta} - \theta_0}{\omega}\Big|\Big)\Big)} \\
&= 2\,\Phi\Big(-\Big|\frac{\hat{\theta} - \theta_0}{\omega}\Big|\Big)
\end{aligned}$$

- p-value: random before the data are plugged in, a statistic.
- **Claim.** Size α:

$$\mathrm{P}(\text{p-value} < \alpha \mid \theta = \theta_0) = \alpha$$

Two-sided p-value under $\theta = \theta_0$

- CDF of $U \sim \text{Uniform}(0,1)$, for $t \in (0,1)$:

$$\mathrm{P}(U \le t) = t$$

- CDF of the p-value, for $t \in (0,1)$:

$$\begin{aligned}
\mathrm{P}(\text{p-value} \le t \mid \theta = \theta_0) &= \mathrm{P}\Big(2\Big(1 - \Phi\Big(\Big| \underbrace{\frac{\hat\theta - \theta_0}{\omega}}_{= Z \sim N(0,1)} \Big|\Big)\Big) \le t \;\Big|\; \theta = \theta_0\Big) \\
&= \mathrm{P}(2(1 - \Phi(|Z|)) \le t) \\
&= \mathrm{P}\Big(1 - \Phi(|Z|) \le \frac{t}{2}\Big) \\
&= \mathrm{P}\Big(\Phi(|Z|) \ge 1 - \frac{t}{2}\Big) \\
&= \mathrm{P}\Big(|Z| \ge \Phi^{-1}\Big(1 - \frac{t}{2}\Big)\Big) \\
&= \mathrm{P}(|Z| \ge z_{1-t/2}) \\
&= \mathrm{P}(Z \ge z_{1-t/2}) + \mathrm{P}(Z \le \underbrace{-z_{1-t/2}}_{= z_{t/2}}) \\
&= \frac{t}{2} + \frac{t}{2} \\
&= t
\end{aligned}$$

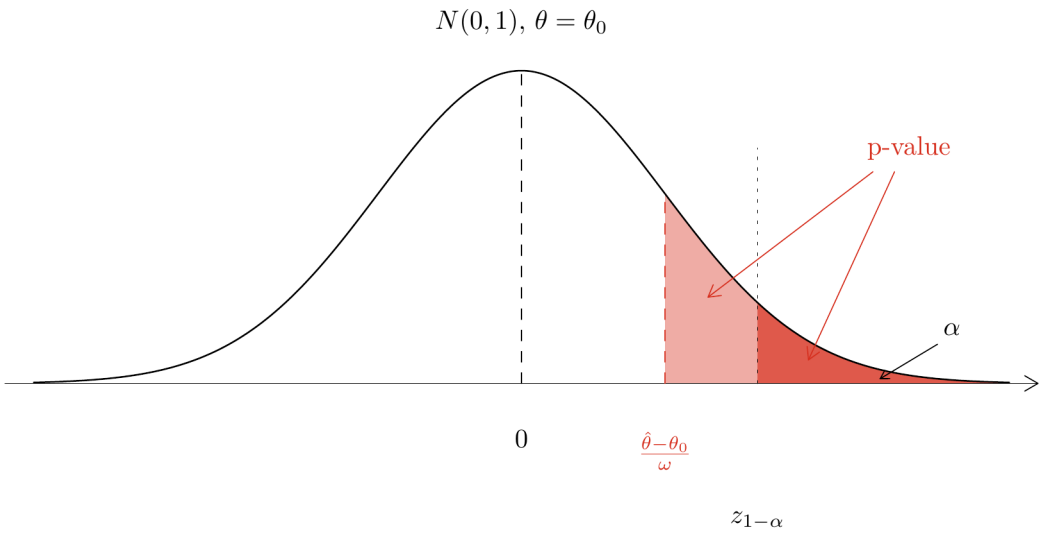
- Under $\theta = \theta_0$, for all $t \in (0,1)$:

$$\mathrm{P}(\text{p-value} \le t \mid \theta = \theta_0) = t$$

$$\implies \boxed{\text{p-value} \sim \text{Uniform}(0,1)}$$

One-sided p-value, right tail

- Right tail beyond $(\hat\theta - \theta_0)/\omega$, for $H_0 : \theta \le \theta_0$:



- Right-tailed test:

$$\boxed{\text{p-value} = 1 - \Phi\Big(\frac{\hat\theta - \theta_0}{\omega}\Big)}$$

- The same, as a probability over $Z^* \sim N(0,1)$ independent of the data:

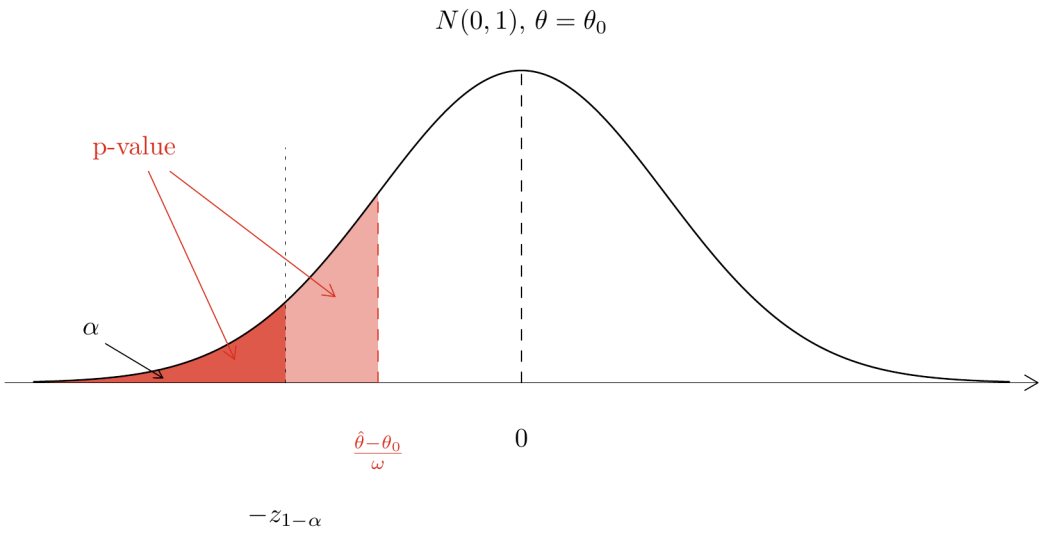
$$\begin{aligned}\text{p-value} &= \mathrm{P}^*\Big(Z^* > \frac{\hat\theta - \theta_0}{\omega} \,\Big|\, \hat\theta\Big) \\ &= 1 - \Phi\Big(\frac{\hat\theta - \theta_0}{\omega}\Big)\end{aligned}$$

- Reject H_0 if p-value $< \alpha$.
- Same decision as the one-sided test:

$$\begin{aligned}& 1 - \Phi\Big(\frac{\hat\theta - \theta_0}{\omega}\Big) < \alpha \\ \iff\quad & \frac{\hat\theta - \theta_0}{\omega} > z_{1-\alpha}\end{aligned}$$

One-sided p-value, left tail

- Left tail below $(\hat\theta - \theta_0)/\omega$, for $H_0 : \theta \geq \theta_0$:



- Left-tailed test: reject if $(\hat\theta - \theta_0)/\omega < -z_{1-\alpha}$.
- p-value:

$$\boxed{\text{p-value} = \Phi\Big(\frac{\hat\theta - \theta_0}{\omega}\Big)}$$

- Reject H_0 if p-value $< \alpha$.

Size of the right-tailed p-value test

- We will use: $\Phi^{-1}(1-\alpha) = z_{1-\alpha}$.

- Supremum over Θ_0, attained at $\theta = \theta_0$:

$$\begin{aligned}
\sup_{\theta \le \theta_0} \mathrm{P}(\text{p-value} < \alpha \mid \theta) &= \sup_{\theta \le \theta_0} \mathrm{P}\Big(1 - \Phi\Big(\frac{\hat{\theta} - \theta_0}{\omega}\Big) < \alpha \;\Big|\; \theta\Big) \\
&= \sup_{\theta \le \theta_0} \mathrm{P}\Big(\Phi\Big(\frac{\hat{\theta} - \theta_0}{\omega}\Big) > 1 - \alpha \;\Big|\; \theta\Big) \\
&= \sup_{\theta \le \theta_0} \mathrm{P}\Big(\Phi\Big(Z + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\le 0}\Big) > 1 - \alpha\Big) \\
&= \mathrm{P}(\Phi(Z) > 1 - \alpha) \\
&= \mathrm{P}(Z > \Phi^{-1}(1 - \alpha)) \\
&= \mathrm{P}(Z > z_{1-\alpha}) \\
&= \alpha
\end{aligned}$$

Distribution of $\Phi(Z)$

- New random variable:

$$W = \Phi(Z), \qquad W \in (\,0, 1\,)$$

- CDF of W:

$$\mathrm{P}(W \le x) = \begin{cases} 0 & x \le 0 \\ ? & 0 < x < 1 \\ 1 & x \ge 1 \end{cases}$$

- We will use: $a \le b \iff g(a) \le g(b)$ for strictly increasing g.
- For $0 < x < 1$, apply Φ^{-1} inside:

$$\begin{aligned}
\mathrm{P}(W \le x) &= \mathrm{P}(\Phi(Z) \le x) \\
&= \mathrm{P}(\Phi^{-1}(\Phi(Z)) \le \Phi^{-1}(x)) \\
&= \mathrm{P}(Z \le \underbrace{\Phi^{-1}(x)}_{= z_x}) \\
&= \Phi(\Phi^{-1}(x)) \\
&= x
\end{aligned}$$

- Hence:

$$\mathrm{P}(\Phi(Z) \le x) = x \quad \Longrightarrow \quad \boxed{\Phi(Z) \sim \mathrm{Uniform}(0, 1)}$$

Distribution of $1 - \Phi(Z)$

- CDF of $1 - \Phi(Z)$, for $0 < u < 1$:

$$\begin{aligned}
F_{1-W}(u) &= \mathrm{P}(1 - W \le u) \\
&= \mathrm{P}(1 - \Phi(Z) \le u) \\
&= \mathrm{P}(\Phi(Z) \ge 1 - u) \\
&= \mathrm{P}(Z \ge \Phi^{-1}(1 - u)) \\
&= 1 - \Phi(\Phi^{-1}(1 - u)) \\
&= 1 - (1 - u) \\
&= u
\end{aligned}$$

- Hence:

$$\mathrm{P}(1 - \Phi(Z) \le u) = u \quad \Longrightarrow \quad \boxed{1 - \Phi(Z) \sim \mathrm{Uniform}(0, 1)}$$

p-values under $\theta = \theta_0$

- $(\hat{\theta} - \theta_0)/\omega = Z$:

$$\begin{aligned} 1 - \Phi(Z) &\sim \text{Uniform}(0,1) \\ \Phi(Z) &\sim \text{Uniform}(0,1) \\ 2(1 - \Phi(|Z|)) &\sim \text{Uniform}(0,1) \end{aligned}$$

- $(\hat{\theta} - \theta_0)/\omega \sim N(0,1)$: critical values $z_{1-\alpha}$, $z_{1-\alpha/2}$ from $N(0,1)$.
- p-value $\sim$ Uniform$(0,1)$: critical value α from Uniform$(0,1)$.
- For every p-value:

$$\begin{aligned} \text{P(p-value} < \alpha \mid \theta = \theta_0) &= \text{P(p-value} \leq \alpha \mid \theta = \theta_0) \\ &= \text{CDF of the p-value at } \alpha, \quad \text{under } \theta = \theta_0 \\ &= \text{P}(V \leq \alpha), \quad V \sim \text{Uniform}(0,1) \\ &= \alpha \end{aligned}$$

- In general: $X \sim F$, F continuous and strictly increasing $\Longrightarrow F(X) \sim$ Uniform$(0,1)$.

Summary

Hypotheses, errors, size

- $\hat{\theta} \sim N(\theta, \omega^2)$, ω^2 known; $\theta \in \Theta = \Theta_0 \cup \Theta_1$, $\Theta_0 \cap \Theta_1 = \emptyset$; $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$.
- Test statistic $T \in S = S_A \cup S_R$; reject H_0 if $T \in S_R$.
- Type I error: reject a true H_0; Type II error: fail to reject a false H_0.
- $S_R \downarrow \Longrightarrow$ P(Type I error) $\downarrow \Longrightarrow$ P(Type II error) $\uparrow$.
- Valid test: $\sup_{\theta \in \Theta_0} \text{P}(T \in S_R \mid \theta) \leq \alpha$; size α test: equality.

Two-sided test and power

- Reject $H_0 : \theta = \theta_0$ if $|(\hat{\theta} - \theta_0)/\omega| > z_{1-\alpha/2}$; at $\alpha = 0.05$, $z_{0.975} \approx 1.96$.
- $|\hat{\theta}/\omega| > z_{1-\alpha/2}$: $\hat{\theta}$ significant at level α.
- Fail to reject $\Longleftrightarrow \theta_0 \in CI_{1-\alpha}$; $CI_{1-\alpha} = \{\theta_0 : T(\theta_0) \leq z_{1-\alpha/2}\}$.
- $(\hat{\theta} - \theta_0)/\omega = Z + (\theta - \theta_0)/\omega \sim N((\theta - \theta_0)/\omega,\ 1)$.
- $\pi(\theta) = \text{P}(|Z + (\theta - \theta_0)/\omega| > z_{1-\alpha/2})$, $\pi(\theta_0) = \alpha$.
- $\sup_{\theta \neq \theta_0} \text{P}(\text{Type II error} \mid \theta) = \sup_{\theta \neq \theta_0} (1 - \pi(\theta)) = 1 - \alpha$.
- Smaller ω: more power.
- Test that ignores the data: size α, power α.

One-sided tests and p-values

- $H_0 : \theta \leq \theta_0$: reject if $(\hat{\theta} - \theta_0)/\omega > z_{1-\alpha}$; $z_{1-\alpha} < z_{1-\alpha/2}$.
- $\sup_{\theta \leq \theta_0} \text{P}(\text{reject } H_0 \mid \theta)$ attained at $\theta = \theta_0$.
- $CI^1_{1-\alpha} = [\hat{\theta} - z_{1-\alpha}\,\omega,\ +\infty)$.
- Against $H_1 : \theta \neq \theta_0$: one-sided test biased; no uniformly most powerful test.

- Unequal tails, $\alpha/3$ right and $2\alpha/3$ left: size α, biased against $H_1 : \theta \neq \theta_0$.
- p-values; reject if p-value $< \alpha$:

$$\begin{aligned} \text{two-sided:} \quad & 2(1 - \Phi(|(\hat{\theta} - \theta_0)/\omega|)) \\ \text{right-tailed:} \quad & 1 - \Phi((\hat{\theta} - \theta_0)/\omega) \\ \text{left-tailed:} \quad & \Phi((\hat{\theta} - \theta_0)/\omega) \end{aligned}$$

- Under $\theta = \theta_0$: every p-value $\sim$ Uniform$(0, 1)$; P(p-value $< \alpha \mid \theta = \theta_0) = \alpha$.